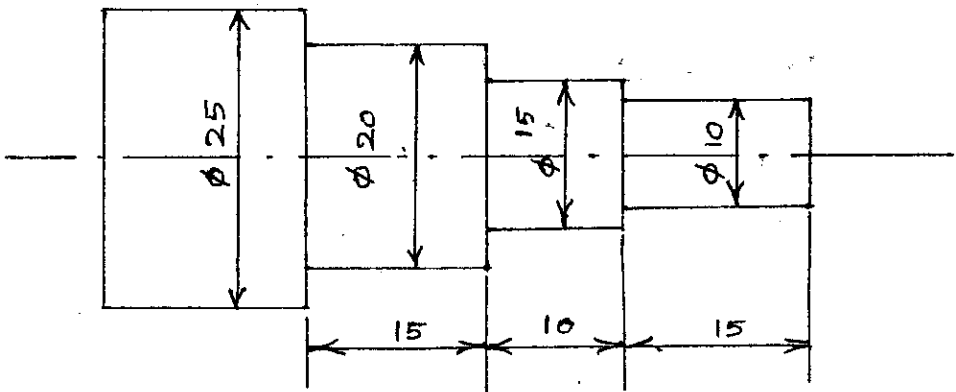


B.Tech Degree VIII Semester Examination, April 2010**EC/EB/EI 804 (C) MECHATRONICS**
(2002 Scheme)

Time: 3 Hours

Maximum Marks: 100

- I. (a) What is mechatronics? Discuss its scope briefly. (10)
 (b) What are the basic components of an NC system? Briefly explain. (15)
OR
- II. (a) Differentiate between NC and CNC. (10)
 (b) With the help of a suitable sketch explain the operation of a re-circulating ball bearing? Why are they used in NC machines? (15)
- III. (a) Discuss the working of stepper motors and servo motors. In which cases are they used in NC machines? Explain. (10)
 (b) Write short notes on the following: (15)
 (i) Encoders (ii) Resolvers (iii) Inductosyn
OR
- IV. (a) What is interpolation? Why is it used in NC machining? Discuss DDA algorithm for linear and circular interpolation. (10)
 (b) Write short notes on the following: (15)
 (i) Flip-Flops (ii) Counters (iii) DAC
- V. (a) Define and explain CIM. (5)
 (b) Briefly discuss the nature and role of the elements of a CIM system. (20)
OR
- VI. (a) Differentiate between 'Manual Programming' and 'Computer Aided Programming' for an NC machine. (10)
 (b) Develop an NC part program for the part given in figure. Assume any missing data.



- VII. (a) Briefly discuss the Common Robot configurations with the help of suitable sketches. (15)
 (b) Write short notes on: (10)
 (i) Robot Control Systems (ii) End effectors
OR
- VIII. (a) Briefly discuss on Robot Programming Languages. (15)
 (b) Write short notes on: (10)
 (i) Basic Robot Motions (DOF) (ii) Robotic Sensors

